Extra Course A: Resonance and Q

Series and parallel RLC at resonance, what Q means, and how to calculate resonant frequency and bandwidth.

Group E5A · 13 pool questions · 30-45 minute presentation · Foundation for Bite #17 · Practice exams at eham.net/exams

Vocabulary First: What Is An "RLC" Circuit?

RLC is shorthand for any circuit that combines all three basic passive components: a **resistor (R)**, an **inductor (L)**, and a **capacitor (C)**. The letters are the schematic-symbol labels (R for resistor, L for inductor (the symbol is commonly attributed to Lenz of Lenz's Law), C for capacitor).

Symbol	What it does	At RF it opposes current with…
R	Burns power as heat. Independent of frequency.	Resistance (in ohms). Same at every frequency.
L	Stores energy in a magnetic field. Resists **changes** in current.	Inductive reactance $X_L = 2\pi fL$. **Bigger** at high frequency.
C	Stores energy in an electric field on two plates. Resists **changes** in voltage.	Capacitive reactance $X_C = 1/(2\pi fC)$. **Smaller** at high frequency.

Inductive reactance and capacitive reactance oppose current the way resistance does, but with a phase shift – and they pull in opposite directions in the math. Sweep the frequency and X_L rises while X_C falls. Somewhere in the middle, they are equal in magnitude but opposite in sign. They cancel. That is resonance.

A **series RLC** circuit puts R, L, and C in a line (current goes through each one). A **parallel RLC** circuit puts L and C across each other in parallel, with R as the lossy element (current splits between the L branch and the C branch). The two topologies behave like mirror images of each other at resonance, which is most of what this course is about.

What "Resonance" Means

Every RLC circuit has a frequency where the inductive reactance (X_L) and the capacitive reactance (X_C) are equal in magnitude. At that frequency they cancel each other, leaving only the circuit's resistance to oppose current. That special frequency is called **resonance**, and the cancellation of opposite reactances is why resonance does interesting things.

Why we care: a resonant circuit lets you build a frequency-selective device with just a coil and a capacitor. Add a resistor (or just use the wire's own loss as R) and you have a tunable filter, an antenna trap, a tank circuit for an oscillator, or the input matching network for a receiver. Once you understand resonance, half the analog parts in a radio stop looking mysterious.

Resonant frequency formula:

$$f = 1 / (2\pi \times \sqrt{(L \times C)})$$

L in henries, C in farads, f in Hz. The L and C combination sets the resonant frequency; the R in the circuit does not.

Series RLC At Resonance

Imagine R, L, and C wired in a single loop with an AC source. Below resonance the capacitor dominates (X_C is large), the impedance is high, and the current is small. Above resonance the inductor dominates (X_L is large), again the impedance is high and the current is small. **At** resonance, X_L and X_C are equal and 180° out of phase with each other – they cancel completely. What is left in the loop is just R. The impedance is at its minimum, the current is at its maximum, and the source sees a purely resistive load.

Counterintuitive consequence: even though X_L and X_C **cancel each other** at the source terminals, the individual voltages across L and C are still very real. With high circulating current and a large reactance, the voltage across the inductor (or the capacitor) by itself can be many times larger than the voltage applied to the whole circuit. They are 180° apart, so they add to zero in the loop equation, but each one individually can be dangerous – a hundred volts in can become a kilovolt across the inductor in a high-Q series resonant tank.

Property	Behavior	Q
Impedance	Drops to a **minimum**, approximately equal to the **circuit resistance R** alone. (E5A03) X_L and X_C cancel; only R is left.	E5A03
Current	Maximum (because Z is minimum and V/Z is at its peak).	—
Voltage across L and C individually	Can be **higher than the applied voltage**. (E5A01) The L and C each see X times the current, and X is much larger than R in a high-Q circuit, so V_L and V_C can vastly exceed $V_{applied}$. (They cancel each other in series, so the total stays right.)	E5A01
Phase of V and I	**In phase.** (E5A08) The reactive parts cancel; the circuit looks purely resistive.	E5A08

Increasing Q	**Internal voltages increase.** (E5A13) Higher Q means higher reactance relative to R, so the V across L and C climbs.	E5A13

Parallel RLC At Resonance

In a parallel RLC, the inductor and capacitor are in parallel with each other (a "tank circuit"), with R representing the loss. The behavior is the mirror image of series resonance.

At resonance, the L branch wants to draw a large current 90° behind the source voltage, and the C branch wants to draw a large current 90° ahead. Those two currents are 180° apart from each other, so they cancel at the input terminals – the input current from the source drops to a minimum, and the input impedance climbs to a maximum (just R). But inside the tank, the L and C are happily trading energy back and forth: when the inductor is collapsing its field, it shoves current into the capacitor; when the capacitor discharges, it shoves current back into the inductor. That sloshing current can be many times larger than what the source supplies. The tank circulates – hence the name. This is the parallel mirror of the "voltage can exceed applied voltage" phenomenon in the series case.

Property	Behavior	Q
Impedance	**Maximum**, approximately equal to the circuit resistance. (E5A04) The L and C currents circulate within the tank and partly cancel at the input terminals.	E5A04
Input current	**Minimum.** (E5A07) Because Z is maximum at resonance.	E5A07
Circulating current inside the tank	**Maximum.** (E5A06) The L and C swap energy back and forth and the current sloshing between them can be many times larger than the input current. This is the parallel-tank version of "voltages can exceed applied voltage."	E5A06

Q: What It Means And How It's Calculated

Q (Quality factor) is a dimensionless number that says how "sharp" a resonance is. High Q = sharp peak, narrow bandwidth, low losses. Low Q = broad peak, wide bandwidth, more losses. Physically, Q is the ratio of energy stored in the L and C versus energy lost in R per cycle: a high-Q tank rings for many cycles before its energy decays away, like a tuning fork that keeps singing; a low-Q tank dies in a few cycles, like a tuning fork wrapped in foam.

Circuit topology	Q formula	Intuition
Series RLC	Q = X / R	Reactance "above" the resistance – bigger gap = sharper.
Parallel RLC	**Q = R / X** (E5A09)	Resistance "above" the reactance – higher parallel-loss R lets the tank ring longer with the same X.

Use either X_L or X_C in the formula; at resonance they are equal so it does not matter which.

Bandwidth From Q

Half-power (3 dB) bandwidth formula:

$$\mathbf{BW = f_0 / Q}$$

where f_0 is the resonant frequency and BW is the bandwidth between the two half-power points (where the response is 3 dB below the peak).

Higher Q in an impedance-matching circuit (like a Pi network or a transmatch) gives a **narrower matching bandwidth**. (E5A05) Good for selectivity, bad for working a whole ham band without retuning.

Worked Examples

Resonant frequency: L = 50 µH, C = 40 pF (E5A02)

$f = 1 / (2\pi \sqrt{(L \times C)}) = 1 / (2\pi \sqrt{(50 \times 10^{-6} \times 40 \times 10^{-12})})$
$\sqrt{(2 \times 10^{-15})} = 4.47 \times 10^{-8}$
$f = 1 / (2\pi \times 4.47 \times 10^{-8}) = 1 / (2.81 \times 10^{-7}) \approx$ **3.56 MHz**

Resonant frequency: L = 50 µH, C = 10 pF (E5A10)

With C reduced by 4×, √C drops by 2×, so f goes up by 2×. From the previous result of 3.56 MHz, f ≈ **7.12 MHz**. (Or compute directly the same way.)

Half-power BW: f_0 = 7.1 MHz, Q = 150 (E5A11)

BW = f_0 / Q = 7,100,000 / 150 ≈ **47.3 kHz**

Half-power BW: f_0 = 3.7 MHz, Q = 118 (E5A12)

BW = 3,700,000 / 118 ≈ **31.4 kHz**

Mental Cheat Sheet

- Resonance: $X_L = X_C$, they cancel, circuit looks resistive.
- $f = 1/(2\pi\sqrt{LC})$. R does not affect resonant frequency.
- Series RLC at resonance: Z = R (minimum), I = max, V across L or C can exceed V applied. V and I in phase.
- Parallel RLC at resonance: Z = R (maximum), input I = min, circulating I in tank = max.
- Q = X/R for series; **Q = R/X for parallel**.
- Higher Q ⟹ narrower bandwidth, higher internal voltages or larger circulating currents.
- BW = f_0 / Q.
- High-Q matching network = narrow matching bandwidth.

Common Traps

- Resonance is NOT determined by R. Only L and C set the frequency.
- Series resonance gives MINIMUM impedance; parallel gives MAXIMUM impedance. Mixing those up is the classic mistake.
- Parallel input current is at a MINIMUM, but circulating current inside the tank is at a MAXIMUM. Read the question carefully – “input current” vs “circulating current.”
- Q for parallel is R/X (the resistance is the loss, kept above the reactance for high Q). Q for series is X/R. Opposite ratios.
- Higher Q means **narrower** bandwidth, not wider. The trap answer is “matching bandwidth is increased.”
- Increasing Q in series RLC raises **internal voltages**, not gain or harmonics.

Takeaway: Resonance is where reactances cancel. Series resonance: Z minimum, I maximum, V across reactances can blow up at high Q. Parallel resonance: Z maximum, input I minimum, circulating I inside the tank can blow up. Q is the sharpness factor: X/R for series, R/X for parallel. Bandwidth = f_0/Q. Higher Q = sharper, narrower, and more dangerous to internal components.

Course A Question Bank (E5A)

E5A01. What can cause the voltage across reactances in a series RLC circuit to be higher than the voltage applied to the entire circuit?
A. Resonance.
B. Capacitance.
C. Low quality factor (Q).
D. Resistance.

E5A02. What is the resonant frequency of an RLC circuit if R is 22 ohms, L is 50 microhenries, and C is 40 picofarads?
A. 44.72 MHz.
B. 22.36 MHz.
C. 3.56 MHz.
D. 1.78 MHz.

E5A03. What is the magnitude of the impedance of a series RLC circuit at resonance?
A. High, compared to the circuit resistance.
B. Approximately equal to capacitive reactance.
C. Approximately equal to inductive reactance.
D. Approximately equal to circuit resistance.

E5A04. What is the magnitude of the impedance of a parallel RLC circuit at resonance?
A. Approximately equal to circuit resistance.
B. Approximately equal to inductive reactance.
C. Low compared to the circuit resistance.
D. High compared to the circuit resistance.

E5A05. What is the result of increasing the Q of an impedance-matching circuit?
A. Matching bandwidth is decreased.
B. Matching bandwidth is increased.
C. Losses increase.
D. Harmonics increase.

E5A06. What is the magnitude of the circulating current within the components of a parallel LC circuit at resonance?
A. It is at a minimum.
B. It is at a maximum.
C. It equals 1 divided by the quantity 2 times pi, times the square root of (inductance L multiplied by capacitance C).
D. It equals 2 times pi, times the square root of (inductance L multiplied by capacitance C).

E5A07. What is the magnitude of the current at the input of a parallel RLC circuit at resonance?
A. Minimum.
B. Maximum.
C. R/L.
D. L/R.

E5A08. What is the phase relationship between the current through and the voltage across a series resonant circuit at resonance?
A. The voltage leads the current by 90 degrees.
B. The current leads the voltage by 90 degrees.
C. The voltage and current are in phase.
D. The voltage and current are 180 degrees out of phase.

E5A09. How is the Q of an RLC parallel resonant circuit calculated?
A. Reactance of either the inductance or capacitance divided by the resistance.
B. Reactance of either the inductance or capacitance multiplied by the resistance.
C. Resistance divided by the reactance of either the inductance or capacitance.
D. Reactance of the inductance multiplied by the reactance of the capacitance.

E5A10. What is the resonant frequency of an RLC circuit if R is 33 ohms, L is 50 microhenries, and C is 10 picofarads?
A. 7.12 MHz.
B. 23.5 kHz.
C. 7.12 kHz.
D. 23.5 MHz.

E5A11. What is the half-power bandwidth of a resonant circuit that has a resonant frequency of 7.1 MHz and a Q of 150?
A. 157.8 Hz.
B. 315.6 Hz.
C. 47.3 kHz.
D. 23.67 kHz.

E5A12. What is the half-power bandwidth of a resonant circuit that has a resonant frequency of 3.7 MHz and a Q of 118?
A. 436.6 kHz.
B. 218.3 kHz.
C. 31.4 kHz.
D. 15.7 kHz.

E5A13. What is an effect of increasing Q in a series resonant circuit?
A. Fewer components are needed for the same performance.
B. Parasitic effects are minimized.
C. Internal voltages increase.
D. Phase shift can become uncontrolled.

Answer Key

E5A01: A	E5A02: C	E5A03: D	E5A04: A
E5A05: A	E5A06: B	E5A07: A	E5A08: C
E5A09: C	E5A10: A	E5A11: C	E5A12: C
E5A13: C			